

PYTHON SET-3

SCENARIO BASED QUESTIONS

1. Scenario: A user is required to enter a valid number in a form, but users sometimes input invalid data.

Write logic to repeatedly prompt the user until they enter a valid integer.

Logic:

1. Ask the user to enter a number.
2. Check whether the input is an integer.
3. If it is valid, accept the number and stop.
4. If it is invalid, display an error message.
5. Ask the user again.
6. Repeat until a valid integer is entered.

2. Scenario: A data analysis tool processes a list of numbers and needs to identify the most frequently occurring value.

Write logic to find the most frequently occurring number in a given list.

Logic:

1. Take the given list of numbers.
2. Check each number in the list.
3. Count how many times each number occurs.
4. Keep track of the number having the highest count.
5. After checking all numbers, return the number with the highest frequency.

3. Scenario: A text-processing application needs to compare words and check if they are anagrams (contain the same letters in a different order).

Write logic to determine whether two given strings are anagrams.

Logic:

1. Take the two strings.
2. Ignore spaces and, if required, ignore differences in uppercase/lowercase.
3. Check whether both strings contain the same number of characters.
4. Count the occurrence of every character in both strings.
5. Compare the character counts.
6. If all counts are the same, they are anagrams.
7. Otherwise, they are not anagrams.

4. Scenario: A speech analysis program needs to count the number of vowel sounds in a given input.

Write logic to count the number of vowels in a given string.

Logic:

1. Take the given string.
2. Examine each character one by one.
3. Check whether the character is a vowel: a, e, i, o, u.
4. If it is a vowel, increase the vowel count by 1.
5. Continue until every character has been checked.
6. Display the final vowel count.

5. Scenario: A text-editing software includes a feature to reverse the order of words in a sentence for stylistic effects.

Write logic to reverse the order of words in a sentence while keeping the words themselves intact.

Logic:

1. Take the complete sentence.
2. Separate the sentence into individual words.
3. Start from the last word.
4. Add the words in reverse order.
5. Keep each word itself unchanged.

6. Scenario: A missing number is detected in a sequence of values stored in a database.

Write logic to find the missing number in a list containing $n-1$ numbers from 1 to n .

Logic:

1. The list should contain numbers from 1 to n .
2. Calculate the expected total of numbers from 1 to n .
3. Calculate the actual total of the numbers present in the list.
4. Subtract the actual total from the expected total.
5. The difference is the missing number.

7. Scenario: An ATM machine processes withdrawal requests and needs to ensure that users cannot withdraw more than their account balance.

Write logic to allow a withdrawal only if the balance is sufficient.

Logic:

1. Take the account balance.
2. Take the amount the user wants to withdraw.
3. Compare the withdrawal amount with the available balance.
4. If the withdrawal amount is less than or equal to the balance, allow the withdrawal.
5. Deduct the withdrawal amount from the balance.
6. If the withdrawal amount is greater than the balance, reject the transaction and show Insufficient Balance.

8. Scenario: A system needs to verify whether a given dataset contains duplicate entries.

Write logic to check whether a given list contains duplicate values.

Logic:

1. Take the given list.
2. Examine each value.
3. Keep track of the values already encountered.

4. When a value appears again, it is a duplicate.
5. If any value is repeated, report that duplicates exist.
6. If no value is repeated, report that all values are unique.

9. Scenario: A digital calculator includes a feature to sum the digits of a number for verification purposes.

Write logic to calculate the sum of all digits in a given integer.

Logic:

1. Take the given integer.
2. Separate the last digit from the number.
3. Add that digit to the total.
4. Remove the last digit from the number.
5. Repeat until there are no digits left.
6. The accumulated total is the sum of the digits.

10. Scenario: A language-learning app wants to verify whether a given sentence is a pangram (contains every letter of the alphabet at least once).

Write logic to check if a given sentence is a pangram.

Logic:

1. Take the given sentence.
2. Ignore spaces, numbers, and punctuation.
3. Convert letters to the same case.
4. Identify all unique alphabet letters present.
5. Check whether all 26 letters from A to Z are present.
6. If every letter occurs at least once, it is a pangram.
7. If even one alphabet letter is missing, it is not a pangram.